Time Series — Solutions

Practice Exam — Week 7: Spectral Density & Spectral Representation

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Problem 1 (5 points)

(a) The ACVS of a real process is real and symmetric, $\gamma_{-\tau} = \gamma_\tau$. Pairing the term at lag τ with the one at lag $-\tau$ in the defining sum,

$$\mathcal{S}(f) = \sum_{\tau\in\mathbb{Z}} \gamma_\tau e^{-2\pi i f\tau} = \gamma_0 + \sum_{\tau\geq 1} \gamma_\tau \big(e^{-2\pi i f\tau} + e^{+2\pi i f\tau}\big)$$
$$= \gamma_0 + 2\sum_{\tau\geq 1} \gamma_\tau \cos(2\pi f\tau),$$

using $e^{-ix} + e^{ix} = 2\cos x$. The right-hand side is a sum of real cosines, hence **real-valued**. It is **even** because cos is even: replacing $f \mapsto -f$ leaves every term unchanged, so $\mathcal{S}(-f) = \mathcal{S}(f)$. (Equivalently, $\mathcal{S}(-f) = \sum_\tau \gamma_\tau e^{2\pi i f\tau} = \sum_\tau \gamma_{-\tau} e^{-2\pi i f\tau} = \mathcal{S}(f)$ after re-indexing $\tau \mapsto -\tau$ and using symmetry.)

(b) Evaluate the inversion formula $\gamma_\tau = \int_{-1/2}^{1/2} \mathcal{S}(f) e^{2\pi i f\tau}\, df$ at lag $\tau = 0$, where the exponential equals 1:

$$\gamma_0 = \int_{-1/2}^{1/2} \mathcal{S}(f)\, e^0\, df = \int_{-1/2}^{1/2} \mathcal{S}(f)\, df.$$

Since $\gamma_0 = \mathrm{Cov}(X_t, X_t) = \mathrm{Var}(X_t)$ (the process has mean 0), this is the spectral form of Parseval: the *total variance* of the process equals the integral of the SDF over the Nyquist band. Because $\mathcal{S} \geq 0$ (it is the transform of a non-negative-definite sequence), $\mathcal{S}(f)\, df$ acts like the "amount of variance/power carried by frequencies in $[f, f + df]$": the SDF **distributes the total power** $\mathrm{Var}(X_t)$ **across frequency**, with peaks marking the bands that dominate the process's fluctuations.

(c) White noise is uncorrelated, so $\gamma_\tau = \sigma^2$ at $\tau = 0$ and $\gamma_\tau = 0$ for $\tau \neq 0$. Only the $\tau = 0$ term survives the sum:

$$\mathcal{S}(f) = \sum_{\tau\in\mathbb{Z}} \gamma_\tau e^{-2\pi i f\tau} = \gamma_0\, e^0 = \boxed{\sigma^2}, \qquad \text{for all } f.$$

The spectrum is **flat**: every frequency carries exactly the same power σ^2, by analogy with white light containing all colours in equal measure — hence "white" noise. (Check: $\int_{-1/2}^{1/2} \sigma^2\, df = \sigma^2 = \gamma_0 = \mathrm{Var}(\varepsilon_t)$, consistent with part (b).)

Problem 2 (5 points)

(a) A summable symmetric sequence $\{\gamma_\tau\}$ is the ACVS of some stationary process **iff it is non-negative definite**, which (in the summable case) is equivalent to its Fourier transform being non-negative everywhere:

$$\mathcal{S}(f) = \sum_{\tau\in\mathbb{Z}} \gamma_\tau e^{-2\pi i f\tau} \;\geq\; 0 \qquad \text{for all } f \in [-\tfrac{1}{2}, \tfrac{1}{2}].$$

This is the right test because, by Herglotz's theorem, every ACVS is non-negative definite and admits an integrated spectrum; in the summable case that spectrum has a density $\mathcal{S}$, and a genuine spectral density — being the limit of the (non-negative) expected periodogram, equivalently $\mathrm{Var}(dZ(f))/df \geq 0$ in the spectral representation — can never be negative. So a single frequency at which $\mathcal{S}(f) < 0$ *disqualifies* the candidate.

(b) The sequence is summable and symmetric, so we may form

$$\mathcal{S}(f) = \sum_{\tau=-K}^{K} e^{-2\pi i\tau f} = 1 + 2\sum_{\tau=1}^{K}\cos(2\pi\tau f) = \frac{\sin\big((2K+1)\pi f\big)}{\sin(\pi f)},$$

the Dirichlet kernel (the closed form follows from summing the geometric series $\sum_{\tau=-K}^{K} e^{-2\pi i\tau f}$). Evaluate at the Nyquist endpoint $f = \frac{1}{2}$, where $\cos(\pi\tau) = (-1)^\tau$. The alternating sum $\sum_{\tau=1}^{K}(-1)^\tau$ equals 0 when K is even and -1 when K is odd, so

$$\mathcal{S}(\tfrac{1}{2}) = 1 + 2\sum_{\tau=1}^{K}(-1)^\tau = \begin{cases} 1, & K \text{ even}, \\ -1, & K \text{ odd}. \end{cases}$$

Thus for every **odd** K (write $K = 2q+1$) we have $\mathcal{S}(\frac{1}{2}) = -1 < 0$, so the SDF goes negative and $\{\gamma_\tau\}$ is **not** a valid ACVS. (For even K this endpoint is harmless, but other frequencies of the Dirichlet kernel still dip below 0, so a flat truncated ACVS is in general not valid — truncating a genuine ACVS breaks non-negative-definiteness.)

(c) This is the ACVS shape of an MA(1). With only lags $0, \pm 1$ nonzero,

$$\mathcal{S}(f) = \gamma_0 + 2\gamma_1\cos(2\pi f) = 1 + 2\rho\cos(2\pi f).$$

Since $\cos(2\pi f)$ ranges over $[-1, 1]$, the minimum of $\mathcal{S}$ is $1 - 2|\rho|$, attained at $f = \frac{1}{2}$ if $\rho > 0$ and at $f = 0$ if $\rho < 0$. Non-negativity $\mathcal{S}(f) \geq 0$ for all f therefore holds iff $1 - 2|\rho| \geq 0$, i.e.

$$\boxed{|\rho| \leq \tfrac{1}{2}}.$$

For $|\rho| > \frac{1}{2}$ the SDF dips below zero and the sequence is *not* a valid ACVS. This matches the known fact that an MA(1), $X_t = \varepsilon_t + \vartheta\varepsilon_{t-1}$, has lag-1 autocorrelation $\rho_1 = \vartheta/(1+\vartheta^2)$, whose modulus is maximised at $|\vartheta| = 1$ giving $|\rho_1| = \frac{1}{2}$: **no** MA(1) (indeed no stationary process at all with this ACVS shape) can have $|\rho_1| > \frac{1}{2}$.

Problem 3 (5 points)

(a) Write $X_t = \sum_{j=0}^{q}\theta_j\varepsilon_{t-j}$ (with $\theta_0 = 1$). Since ε is mean-zero, so is X_t, and by bilinearity of covariance together with $\mathbb{E}[\varepsilon_{t-j}\varepsilon_{t+\tau-k}] = \sigma^2\mathbf{1}_{\{t-j=t+\tau-k\}} = \sigma^2\mathbf{1}_{\{k-j=\tau\}}$,

$$\gamma_\tau = \mathrm{Cov}(X_t, X_{t+\tau}) = \sum_{j=0}^{q}\sum_{k=0}^{q}\theta_j\theta_k\,\mathbb{E}[\varepsilon_{t-j}\varepsilon_{t+\tau-k}] = \sigma^2\sum_{j=0}^{q}\sum_{k=0}^{q}\theta_j\theta_k\,\mathbf{1}_{\{\tau=k-j\}}.$$

Collapsing the indicator (set $k = j+\tau$) gives the single-sum form

$$\gamma_\tau = \sigma^2\sum_{j=0}^{q}\theta_j\theta_{j+|\tau|} \quad (|\tau| \leq q), \qquad \gamma_\tau = 0 \ (|\tau| > q),$$

where we used $\theta_j = 0$ for $j > q$; the value depends only on $|\tau|$ because the double sum is symmetric in $j \leftrightarrow k$. So the ACVS is finitely supported on $|\tau| \leq q$, hence trivially in ℓ^1 and the SDF exists.

(b) Transform the ACVS term by term, keeping the double-sum form so the indicator selects

$\tau = k - j$:

$$\begin{aligned}
\mathcal{S}(f) = \sum_{\tau\in\mathbb{Z}} \gamma_\tau e^{-2\pi i\tau f} &= \sigma^2 \sum_{\tau\in\mathbb{Z}} \sum_{j=0}^{q} \sum_{k=0}^{q} \theta_j\theta_k\, \mathbf{1}_{\{\tau=k-j\}}\, e^{-2\pi i\tau f} \\
&= \sigma^2 \sum_{j=0}^{q} \sum_{k=0}^{q} \theta_j\theta_k\, e^{-2\pi i(k-j)f} \\
&= \sigma^2 \Big(\sum_{j=0}^{q} \theta_j e^{+2\pi ijf}\Big)\Big(\sum_{k=0}^{q} \theta_k e^{-2\pi ikf}\Big) \\
&= \sigma^2\, \overline{\Big(\sum_{k=0}^{q} \theta_k e^{-2\pi ikf}\Big)}\Big(\sum_{k=0}^{q} \theta_k e^{-2\pi ikf}\Big) = \boxed{\sigma^2 \Big|\sum_{j=0}^{q} \theta_j e^{-2\pi ijf}\Big|^2}.
\end{aligned}$$

In the third line the sum over τ collapsed the indicator, and $e^{-2\pi i(k-j)f} = e^{+2\pi ijf}\, e^{-2\pi ikf}$ factored the double sum into a number times its conjugate (the θ_j are real). The SDF is the squared modulus of the MA transfer function $\Theta(e^{-2\pi if}) = \sum_j \theta_j e^{-2\pi ijf}$; **a squared modulus is ≥ 0 automatically**, so this form makes the necessary non-negativity of an SDF manifest.

(c) For the MA(1), $\Theta(z) = 1 - \theta z$, so $\sum_j \theta_j e^{-2\pi ijf} = 1 - \theta e^{-2\pi if}$ and

$$\mathcal{S}(f) = \sigma^2\big|1 - \theta e^{-2\pi if}\big|^2 = \sigma^2\big(1 - \theta e^{-2\pi if}\big)\big(1 - \theta e^{+2\pi if}\big) = \sigma^2\big(1 + \theta^2 - 2\theta\cos(2\pi f)\big),$$

matching the stated form (and the direct computation from $\gamma_0 = \sigma^2(1+\theta^2)$, $\gamma_{\pm1} = -\sigma^2\theta$). For $\theta > 0$ the term $-2\theta\cos(2\pi f)$ is smallest (most negative) when $\cos(2\pi f) = +1$, i.e. at $f = 0$, and largest when $\cos(2\pi f) = -1$, i.e. at $f = \frac{1}{2}$. Hence

$$\min_f \mathcal{S} = \mathcal{S}(0) = \sigma^2(1 + \theta^2 - 2\theta) = \sigma^2(1-\theta)^2 \quad (\text{at } f = 0),$$

$$\max_f \mathcal{S} = \mathcal{S}(\tfrac{1}{2}) = \sigma^2(1 + \theta^2 + 2\theta) = \sigma^2(1+\theta)^2 \quad (\text{at } f = \tfrac{1}{2}).$$

So a positive θ gives a **high-pass** shape: power is suppressed at low frequency ($f = 0$) and amplified at the Nyquist frequency ($f = \frac{1}{2}$). (For $\theta < 0$ the two extremes swap, giving a low-pass spectrum; and $\mathcal{S} \geq 0$ throughout, touching 0 at one frequency only when $|\theta| = 1$.)

Problem 4 (5 points)

(a) By definition $U(f) = \int_{-\infty}^{\infty} u(x)e^{-2\pi ixf}\,dx$ with $u(x) = \int g(x-y)h(y)\,dy$. Substitute and exchange the order of integration:

$$\begin{aligned}
U(f) &= \int_{-\infty}^{\infty}\Big(\int_{-\infty}^{\infty} g(x-y)h(y)\,dy\Big)e^{-2\pi ixf}\,dx \\
&= \int_{-\infty}^{\infty}\Big(\int_{-\infty}^{\infty} g(x-y)e^{-2\pi ixf}\,dx\Big)h(y)\,dy \qquad (\text{Fubini}) \\
&= \int_{-\infty}^{\infty}\Big(\int_{-\infty}^{\infty} g(s)e^{-2\pi isf}\,ds\Big)e^{-2\pi iyf}\,h(y)\,dy \quad (s = x - y) \\
&= G(f)\int_{-\infty}^{\infty} h(y)e^{-2\pi iyf}\,dy = G(f)\,H(f).
\end{aligned}$$

In the third line the substitution $s = x - y$ (for fixed y) splits the kernel as $e^{-2\pi ixf} = e^{-2\pi isf}e^{-2\pi iyf}$, making the inner integral exactly $G(f)$, independent of y. The interchange of the two integrals is legitimate by **Fubini's theorem**, justified because the integrand is absolutely integrable:

$$\iint_{\mathbb{R}^2} \big|g(x-y)h(y)\big|\,dx\,dy = \int_{\mathbb{R}} |h(y)|\Big(\int_{\mathbb{R}} |g(s)|\,ds\Big)dy = \|g\|_1\,\|h\|_1 < \infty,$$

since $g, h \in L^1$. Hence $u = g*h \iff U = G \cdot H$: convolution in time is multiplication in frequency.

(b) For $t \in \Delta\mathbb{Z}$ both inversion formulas give the *same* value $g(t)$ — the continuous one from $\mathcal{G}$, and the discrete-time (Fourier-series) one from G on one period:

$$g(t) = \int_{-\infty}^{\infty} \mathcal{G}(f) e^{2\pi i f t}\, df, \qquad g(t) = \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f) e^{2\pi i f t}\, df.$$

Split the real line in the first integral into period-length blocks centred at k/Δ and shift each block by $-k/\Delta$:

$$\begin{aligned} g(t) &= \sum_{k\in\mathbb{Z}} \int_{k/\Delta-1/(2\Delta)}^{k/\Delta+1/(2\Delta)} \mathcal{G}(f) e^{2\pi i f t}\, df = \sum_{k\in\mathbb{Z}} \int_{-1/(2\Delta)}^{1/(2\Delta)} \mathcal{G}(f+k/\Delta)\, e^{2\pi i (f+k/\Delta)t}\, df \\ &= \sum_{k\in\mathbb{Z}} \int_{-1/(2\Delta)}^{1/(2\Delta)} \mathcal{G}(f+k/\Delta)\, e^{2\pi i f t}\, df, \end{aligned}$$

because $t = z\Delta$ for some $z \in \mathbb{Z}$ makes the extra phase $e^{2\pi i (k/\Delta)t} = e^{2\pi i k z} = 1$. Under the summability hypothesis $\sum_k \int |\mathcal{G}(f+k/\Delta)|\, df < \infty$ we may swap sum and integral:

$$\int_{-1/(2\Delta)}^{1/(2\Delta)} \Big(\sum_{k\in\mathbb{Z}} \mathcal{G}(f+k/\Delta)\Big) e^{2\pi i f t}\, df = g(t) = \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f) e^{2\pi i f t}\, df.$$

Equality of these Fourier-series coefficients for all $t \in \Delta\mathbb{Z}$ forces $G(f) = \sum_{k\in\mathbb{Z}} \mathcal{G}(f+k/\Delta)$ for a.e. f (and for all f, both sides being continuous). Sampling thus **folds** all the continuous content at the frequencies $\{f + k/\Delta\}$ onto the single frequency f. The **Nyquist frequency** is $f_N = 1/(2\Delta)$ (one half the sampling rate): content beyond $\pm f_N$ cannot be recovered from the samples.

(c) With $\Delta = 1$ s the Nyquist band is $[-\frac{1}{2}, \frac{1}{2}]$ Hz and the aliases of $f_0 = 0.8$ Hz are $f_0 + k = 0.8 + k$, $k \in \mathbb{Z}$. The one landing inside $[-\frac{1}{2}, \frac{1}{2}]$ is

$$0.8 + (-1) = -0.2 \text{ Hz},$$

so $f_0 = 0.8$ Hz **folds to** -0.2 Hz, i.e. appears at $|-0.2| = 0.2$ Hz (a cosine has a symmetric line pair, so it is observed as a 0.2 Hz oscillation — the wagon-wheel effect). To resolve f_0 without aliasing one needs the Nyquist frequency above f_0: $1/(2\Delta) > 0.8$, i.e. a **sampling rate** $1/\Delta > 1.6$ **Hz** (any rate strictly above 1.6 samples per second).